SPE-227715-MS

Effective Foaming Agents to Rejuvenate Gas Wells with Downhole Liquid Buildup Issues in High Temperature and Salinity Reservoirs

Chang Da, Beijing Research Center, Aramco Asia, Beijing, PR China; Prasad Karadkar, Sharefah Alsultan, and Mohammed Alotaibi, EXPEC Advanced Research Center, Saudi Aramco, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

This study aims to develop effective foaming agents that can be applied for liquid unloading of gas wells under high temperature and high salinity (HTHS) conditions. Up to nine commercial products from the local market in China were evaluated. Among them, the formulated foaming agent AFA-F was selected due to its good compatibility with HTHS condition (up to 150°C and salinity of 15,6000 ppm), along with strong foaming ability and great stability even after incubation at 150°C for up to 16 hours. The chemical stability was confirmed through TGA with no degradation below 220°C. Moreover, the liquid carrying capacity was more than 90% using the selected foaming agent AFA-F at a relatively low concentration. Even in the presence of condensate oil up to 80%, the liquid carrying capacity could still be maintained up to 75%. The high performance of this foaming agent indicated great potential for liquid unloading applications for gas wells with HTHS conditions.

Introduction

Liquid buildup in the wellbore is one of the major causes of production decline in gas wells, in which case additional energy is needed to drain off the liquid to solve this problem (Sankar and Karthi 2019). Foaming agents offer a method to reduce the liquid density to make it easier to be lifted with the gas flow, unloading the accumulated liquid in gas wells (Wills et al. 2008; Yang et al 2013; Wu et al. 2020). The main ingredient of foaming agents is surfactants. The objective of this proposal is to develop foaming agents for liquid unloading applications in gas wells under HTHS conditions. Liquid unloading is a procedure to remove the liquids that have accumulated in a gas well to surface. The liquids accumulation can include brine, oil, and condensate, induced by various factors including decreases of gas velocity in the well, decreases of reservoir pressure, or changing gas to liquid ratios. As liquids accumulate, production of natural gas can be significantly inhibited, which makes the liquids unloading from gas wells so necessary and urgent to restore well production.

Liquids can be removed from well bores in a variety of methods (Tugan wt al. 2020; Tan et al. 2023). For example, the well tubing can be modified to increase gas velocity or a pump may be installed to remove downhole liquids (Veeken et al. 2011). One method that has been recently popular and widely-used is to remove liquids by adding foaming agents (such as surfactants or nanoparticles) to generate foams with strong lifting capacity (Ajani et al. 2016; Junwen et al. 2016). Foam can reduce the density and surface tension of the fluids trapped downhole, which reduces the critical gas velocity needed to lift fluids to surface and thus aids liquid removal (Lea et al. 2004). Compared to other liquid unloading methods, the addition of foaming agents is one of the most economic applications, from the aspects of both cost and energy.

Although the studies on foaming agents are abundant, successful candidates suitable for field applications under high temperature and high salinity conditions have rarely been reported. The HTHS conditions have significantly limited the applications of various types of surfactants. For example, most nonionic (and some anionic) surfactants may demonstrate poor temperature tolerance (due to cloud-point issues) and mediocre salt tolerance. Moreover, another challenge comes as the natural oil/condensate may reach very high ratio in the liquids trapped downhole, thus resulting in severely rupture of foams generated by the foaming agents, and inhibiting the overall capacity of foaming and liquid carrying. In this study, the goal was to develop and screen for commercially-available and cost-effective foaming agents from the Chinese local market, that show good performance in the liquid unloading applications for gas wells under HTHS conditions.

Experimental

Chemicals

NaCl, KCl, $MgCl_2 \cdot 6H_2O$, $CaCl_2 \cdot 2H_2O$, $NaHCO_3$ and Na_2SO_4 were purchased from Aladdin Chemicals (Shanghai, China). The high salinity water (HSW) with salinity up to 15.6% was used in this study, and prepared according to the composition shown in the Table 1, then filtrated the exceeded salt with 0.22 µm membrane after incubation at 95°C. For the foaming agents, formulated products AFA-F and AFA-Q were both from Ningbo Fengcheng. Cationic surfactant cetyltrimethylammonium bromide (CTAB) was from Sinopharm (SCR). Anionic surfactant sodium dodecyl sulfate (SDS) was from Sinopharm (SCR), and C12-14 alpha olefin sulfonate (AOS1214) was from Mackun. Nonionic emulsifier OP-10 was from Guangdong Zhonglianbang, and surfactant alkyl polyglycoside (APG) was from BASF. Zwitterionic surfactants C12-14 amidopropyl sulfobetaine (HSB1214) and C16-18 amidopropyl sulfobetaine (HSB1618) were from Usolf.

Table 1—Compositions of high salinity water (HSW) used in this study, with salinity up to 156,000 total-dissolved-solids

Salt	Salinity (g/L)
NaCl	132.28
$MgCl_2 \cdot 6H_2O$	5.20
$CaCl_2 \cdot 2H_2O$	28.59
Na_2SO_4	0.51
$NaHCO_3$	0.25

Compatibility Tests in HSW at Elevated Temperatures

The compatibility of the foaming agents was preliminarily evaluated in HSW at a constant concentration of 1 wt% (10,000 ppm) at elevated temperature. The foaming agent candidates were dissolved in HSW, loaded in sealed glass vials and incubated in oven at temperature up to 95°C overnight, and the results were determined through visual observation. The recorded observation was then categorized into three types: clear and homogeneous (*H*), cloudy (*C*) or poor with precipitation (*P*). *H* represents homogeneous aqueous phase in clear or white color; *C* stands for very minor cloudiness in aqueous phase, mostly homogeneous aqueous phase; and *P* means major phase separation observed on the glass wall or at the bottom of the vial.

Bulk Foam Tests

Foamability and stability of bulk foams were studied through a Dynamic Foam Analyzer DFA-100 (Kruss, Germany) at both room temperature (25 °C) and up to 80 °C. The solution of up to 20,000 ppm foaming agents in HSW was loaded with 50 mL into the glass cylinder and nitrogen was bubbled into the solution at a constant flow rate of 0.1 mL/min for 1 minute. Then the foam height was initially scanned and lively recorded for up to 60 minutes, and the drainage time with 25% and 50% decayed foam volume was recorded as the criteria for foam stability. The measurements were repeated for up to 3 times to ensure reproducibility, and the final results were averaged from the repeated measurements.

Moreover, the foam stability was further evaluated to guarantee effectiveness and performance under extremely high temperatures close to the downhole condition in targeted gas wells. The solutions of foaming agents in HSW were incubated at temperatures up to 150°C for up to 16 hours, sealed in Penicillin bottles to prevent leakage under elevated temperatures above boiling point. The foaming performance and foam stability were then evaluated and compared before the aging of the samples.

Visualized Liquid Carrying Tests through Nitrogen Bubbling

Solution of foaming agents with various concentrations in HSW was first loaded in the visualized cell for up to 200 mL. Then nitrogen was blown from the bottom of the cell and bubbled into the solution at a constant pressure of 10 psi for up to 5 minutes. Foam was generated during the gas bubbling, as shown in Figure 1, and then transferred from the top of the visualized cell into a gradual cylinder as the liquid collector. After the bubbling process completed, volume of solution collected in the cylinder was recorded and the liquid carrying ratio was calculated, based on the removed liquid volume over total liquid volume at the beginning.

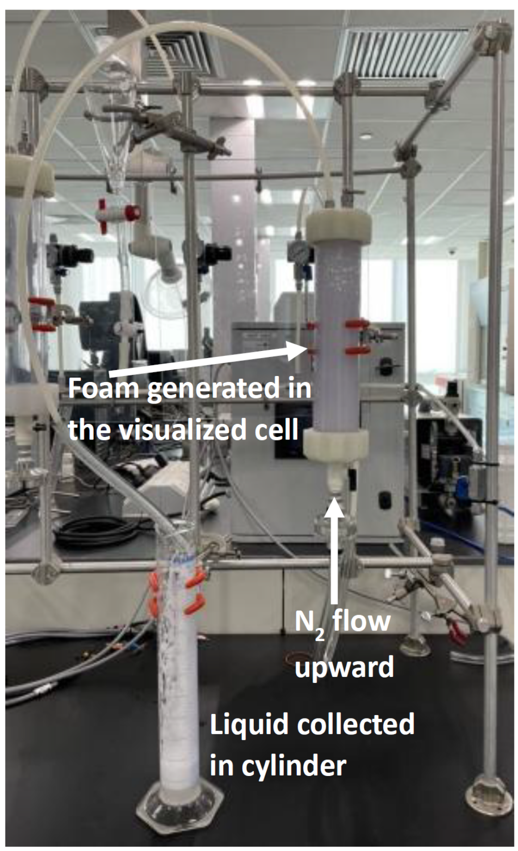

Figure 1—In-house developed setup for the visualized liquid carrying tests.

Moreover, condensate oil up to 80% was also added to the solution in the visualized cell before bubbling as the test of oil effect on liquid carrying capacity of foams. The total liquid volume was maintained as 200 mL again, with up to 160 mL as condensate oil. This step was to evaluate if the high performance of foams could be maintained when in contact with a high fraction of oil/condensate in the trapped liquids for real gas wells. Again the nitrogen was blown from the bottom of the cell and bubbled into the solution at a constant pressure of 10 psi for up to 5 minutes. And the removed liquid volume was recorded in the cylinder.

Chemical Stability Tests at Elevated Temperatures

The thermal stability of the foaming agents at elevated temperatures was verified through Thermogravimetric Analysis (TGA). The sample of the foaming agent (containing water) was first dried in oven and prepared in the cuvette, with the initial mass weighted after sufficient drying process. Then the sample was heated in the TGA apparatus from 40°C all the way up to 1000°C, while the weighted mass was recorded along with temperature increase, to analyze detailed degradation process.

Results and Discussion

Compatibility under HTHS Condition

The first group of tests were designed to preliminarily screen for foaming agents compatible with HTHS conditions under real gas wells. All the samples collected from local Chinese markets were added to HSW in concentration as high as 10,000 ppm, and then incubated in oven at 95°C overnight (up to 16 hours). The results were obtained through visualized observation and shown in Table 2. Obviously, the selected anionic and nonionic surfactants showed poorer tolerance against high temperature and salinity condition, compared to the chosed cationic and zwitterionic ones, which matched up with examples previously reported in literature. Thus, further evaluation would be focused on those with prioritized compatibility, specifically AFA-F, CTAB, and HSB1618.

Table 2—Visualized observations of foaming agents in HSW after at 95°C overnight (*H* stands for clear and homogeneous aqueous phase, *C* stands for minor cloudiness, and *P* stands for severe precipitation)

Foaming agent	Type	Compatibility
AFA-F	Formulated	*H*
AFA-Q		*C*
CTAB	Cationic	*H*
SDS	Anionic	*P*
AOS1416		*P*
OP-10	Nonionic	*C*
APG		*C*
HSB1214	Zwitterionic	*H*
HSB1618		*H*

Foamability and Stability of Bulk Foams

With selected foaming agents demonstrating good compatibility and strong tolerance against HTHS condition, performances on bulk foams were further studied, specifically foam generation and stability, under both 25 °C and 80 °C. With 20,000 ppm of various foaming agents prepared in 40 mL HSW, the foaming process lasted for 60s with 0.1 L/min N_2 bubbled into the solution in the DFA apparatus, reaching up to the foam quality of approximately 70% before the foam decaying. Such foam quality was controlled to match up the commonly-used foams in the scenarios of oilfield applications.

The results were shown in Table 3 and 4. Clearly, at room temperature, the formulated product of AFA-F showed very similar maximum foam volume compared to the zwitterionic HSB1618 and cationic CTAB after the foaming process, thus the 3 foaming agents showed comparable foamability. However, the foam in presence of the formulated foaming agent AFA-F was much more stable compared to the cases with the other 2 single surfactants, demonstrated as significantly longer drainage times compared to the other two.

Table 3—Foaming ability and stability of selected foaming agents at room temperature (25 °C).

Foaming Agent	Maximum Foam Volume (mL)	25% Drainage time (s)	50% Drainage time (s)
AFA-F	161	120	145
HSB1618	160	80	90
CTAB	159	110	130

Table 4—Foaming ability and stability of selected foaming agents at elevated temperature (80 °C).

Foaming Agent	Maximum Foam Volume (mL)	25% Drainage time (s)	50% Drainage time (s)
AFA-F	162	200	220
HSB1618	163	130	145
CTAB	139	150	160

For the foams at 80 °C, clearly the initial foam height with CTAB was > 10% lower than the cases with the other two, suggesting the foamability of CTAB at elevated temperatures became less efficient. For the foam stability, similar phenomena as at room temperature were also observed at 80 °C, as the improvement of AFA-F on foam stability was more significant at elevated temperatures. Thus, it could be reasonably estimated that at even higher temperatures, the enhanced foam stability in presence of AFA-F would be more effective and outstanding compared to the other surfactants. Thus the formulated foaming agent AFA-F was proceeded to further evaluations, including thermal stability and liquid carrying capacity.

Thermal Resistance and Chemical Stability at Elevated Temperatures

Since the formulated sample AFA-F was selected with the outstanding foamability and foam stability, further evaluations was conducted to be focus on this specific foaming agent. The foaming performance of 20,000 ppm AFA-F in HSW before and after incubation at 150 °C overnight was then verified, as the severe conditions in the downhole gas wells could easily go beyond 80 °C in the previous tests. As shown in Table 5, the foaming performance was not affected after incubation at 150 °C for up to 16 hours. Both the maximum foam volume and drainage time were very much comparable before and after the incubation, suggesting maintained foam strength and stability at extremely high temperature. The comparisons of foam height and bubble size vs time were also shown in Figure 2, with curves demonstrating both before and after incubation at 150 °C. Both the foam height decay and bubble size expansion were not influenced by the high temperature, with mostly overlapping curve in Figure 2a and b. Thus, the AFA-F was proved with strong tolerance against conditions with severely high temperature and salinity, making it a suitable candidate for gas well applications.

Table 5—Foaming performance of 20000 ppm AFA-F in HSW before and after aging at 150 °C for 16 hours

	AFA-F	Maximum Foam Volume (mL)	25% Drainage time (s)	50% Drainage time (s)
25°C	Before aging	150.6	160.0	215.0
	After aging	151.9	169.8	219.8
80°C	Before aging	163.5	180.0	200.0
	After aging	164.4	174.9	194.9

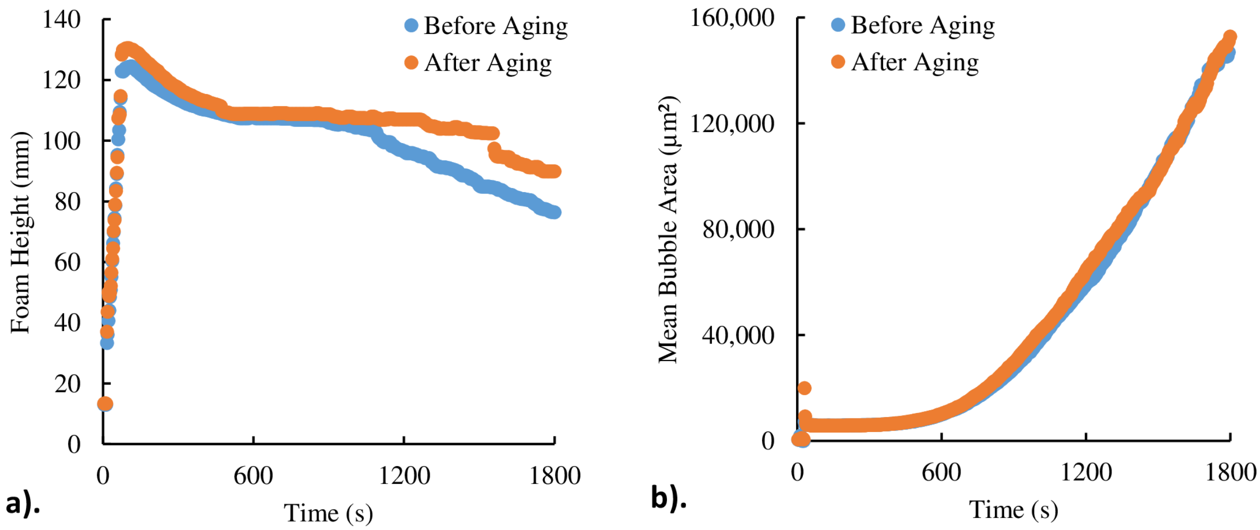

Figure 2—The change of a). foam height and b). bubble size over time of foam generated with 0.1 L/min N_2 bubbling for 60 s into 20000 ppm AFA-F in of 40 mL HSW, comparison between before and after aging at 150°C for 16 hours and analyzed through DFA at 25°C.

Moreover, the chemical stability of the foaming agent at elevated temperatures was evaluated to determine the threshold temperature at which the selected foaming agent AFA-F will start degrading, and thus to avoid injecting degraded products into the gas wells and ensure operational safety. The sample was analyzed through TGA, after most of the water in the foaming agent product had been removed. The results were shown in Figure 3. First of all, very minor mass reduction was observed while the temperature was maintained before the temperature reached 200°C, possibly due to the evaporation of residual water left in the sample. The major mass loss was observed starting when the temperature reached above 220°C, indicating the degradation of the active components initiated. The results verified the very stable chemical structure of the formulated foaming agent AFA-F under 200°C, which was definitely suitable for targeted gas wells with downhole temperatures up to 150°C.

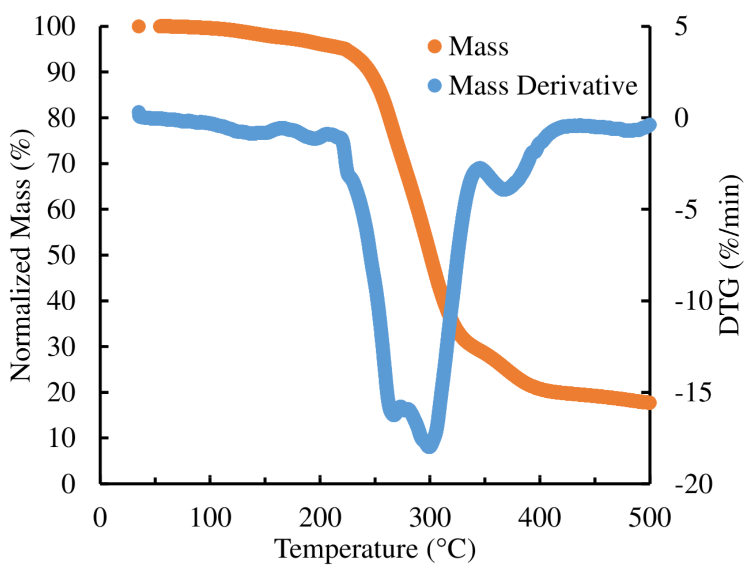

Figure 3—Results of TGA analysis on water-based foaming agent AFA-F.

Liquid Carrying Capacity in Presence of Condensate Oil

Further characterizations were conducted with AFA-F on liquid carrying capacity, to evaluate its performance under real downhole scenarios of liquid unloading applications. Basically, the ratio between the volume of liquid loaded and the volume carried out was recorded, in presence of various concentrations of AFA-F. As shown in Table 6, the liquid carrying capacity of nitrogen only without foaming agents was relatively low under all scenarios, as it never exceeds 15% under all water to oil ratios. However, with 20,000 ppm AFA-F added, the liquid carrying capacity became incredibly good in pure aqueous scenario, recovered up to 96% of the loaded liquid. To mimic the scenario of high content of oil/condensate in the downhole trapped liquids, up to 80% condensate was added to the loaded liquid. The presence of oil slightly dampened the liquid carrying capacity of AFA-F, but the results could be still considered satisfactory, with >75% of liquid can be removed within only 5 min of N_2 bubbling in presence condensate oil up to 80% in volume. Such strong performance of liquid carrying indicated great potential of the selected foaming agent AFA-F for real liquid unloading applications in downhole gas wells.

Table 6—Liquid carry capacity of AFA-F under various condensate ratio in accumulated liquids.

Water : Oil	Liquid Carrying Capacity	
	No foaming agent	2% foaming agent
10:0	<10%	96%
8:2	<10%	90%
5:5	10%	82%
2:8	15%	75%

Conclusions and Recommendation

In this study, out of 9 commercially-available foaming agents from the Chinese market were screened for the best performance. The selected foaming agent AFA-F showed good compatibility with HTHS condition, up to 150°C and 156,000 ppm TDS. Foaming tests at temperature up to 80°C showed great foaming ability and stability with AFA-F, even after aged at 150°C for up to 16h overnight. The liquid carrying capacity reached > 95% with the selected foaming agent AFA-F, and was maintained as >75% when in presence of 80% condensate oil. The significantly good performance of this foaming agent indicated great potential for liquid unloading applications at HTHS gas well scenario.

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